

## Days 1–10: Java Basics (Foundation)

### Day 1: Setup + Java Introduction

- Install JDK
- JVM, JRE, JDK
- First Java program
- Compilation and execution flow
- main() method

Practice:

- Print statements
  - Basic programs
- 

## Day 1 - Java Introduction (Complete Guide)

### Step 1: Programming ante enti?

Manam friends ki instructions istam kada.

Example:

Harika, kitchen ki velli,  
Bottle thechuko,  
Water fill cheyi,  
Room ki ra.

Idi instructions.

Computer ki kuda alane instructions ivvali.

Computer mana language ardham cheskodu.

Computer ki only

0 and 1

(binary)

ardham.

Kabatti manam oka language use chestam.

Examples:

- Java
- Python

- C
- C++
- JavaScript

Java lo instructions rayadam ni

## **Programming**

antaru.

---

### **Real Life Example**

Teacher cheptaru

Take attendance

Student attendance teesukuntadu.

Computer ki

Print Hello

ani cheppali.

Computer

Hello

ani print chestundi.

---

### **Why Java?**

Java is one of the most popular programming languages.

Enduku?

Because

✓ Easy

✓ Secure

✓ Platform Independent: oka sari code rasi ekkadaina use chesukovacchu

✓ Object Oriented: everything is object

✓ Used Everywhere

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Companies using Java

- Google
- Amazon
- Flipkart
- Netflix
- Uber
- Banking Systems
- Android Apps (older versions)
- Enterprise Software

Almost every company uses Java somewhere.

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## **Java History**

Java ni

### **James Gosling**

1995 lo create chesaru.

Goal:

Write Once

Run Anywhere

Idi Java biggest advantage.

Meaning

One system lo code rayachu.

Windows

↓

Linux

↓

Mac

Anni places lo same code run avutundi.

---

## **Why Name Java?**

Initially

Oak

ani name.

Already Oak ane company undi.

Kabatti coffee tagutunnappudu

Java Coffee

kanipinchindi.

Name change chesaru

Java

---

## **Java Architecture**

Idi interview lo compulsory.

Before understanding Java

Meeru 3 words ardham cheskovali.

JDK

JRE

JVM

Students maximum ikkade confuse avtaru.

Let's understand with one example.

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## **Imagine You Want To Cook Biryani**

Need

Rice

↓

Vegetables

↓

Stove

↓

Gas

↓

Utensils

↓

Cook

Everything kavali.

Only rice unte biryani avvadu.

Same Java lo kuda.

---

## **Java Program Run Avvali Ante**

Need

JDK

↓

JRE

↓

JVM

Let's understand one by one.

---

## **What is JVM?**

Full Form

Java Virtual Machine

Very Important.

Interview Question.

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Imagine

Computer oka Telugu person.

Java program oka English person.

Telugu person ki English raadhu.

Need translator.

Translator



English



Telugu

Translate chestadu.

Same

Java Program



JVM



Machine Language

JVM translates Java bytecode into machine code that the computer can execute.

Without JVM

Java Program run avvadu.

---

## **JVM Work**

Java Code



Compiler



Bytecode



JVM



Machine Code



Output

---

### **JVM Responsibilities**

JVM chese panulu

Memory allocate chestundi

Program run chestundi

Objects create chestundi

Garbage collect chestundi

Errors detect chestundi

---

### **Garbage Collection**

Suppose

Phone storage full.

Delete unnecessary files.

Same

Java lo unused objects delete chestundi.

Automatic ga.

Programmer delete cheyyalsina avasaram ledi.

Idi Java advantage.

---

### **What is JRE?**

Full Form

Java Runtime Environment

Simple ga

Program run cheyyadaniki kavalsina environment.

Contains

JVM

+

Libraries

Diagram

JRE

|

|---- JVM

|

|---- Libraries

Libraries ante

Already ready-made code.

Example

Math

Date

Scanner

String

Collections

Etc.

---

### Real Life Example

JVM

↓

Engine

Libraries

↓

Fuel



JRE

↓

Complete Car

---

### **What is JDK?**

Full Form

Java Development Kit

Developer ki kavalsina complete package.

Contains

Compiler

JRE

JVM

Debugging Tools

Development Tools

Diagram

JDK

|

|---- Compiler

|

|---- JRE

|

|----- JVM

|

|----- Libraries

|

|---- Tools

---

## Easy Trick

JDK

↓

JRE

↓

JVM

Remember

Big Box

↓

Medium Box

↓

Small Box

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### **One More Real Life Example**

Restaurant

Owner

↓

Kitchen

↓

Chef

Owner = JDK

Kitchen = JRE

Chef = JVM

Without chef

Food prepare avvadu.

Without kitchen

Chef work cheyyaledu.

Without owner

Restaurant ledu.

---

### **Installation**

Google

Oracle JDK Download

or

OpenJDK

Download latest LTS version (e.g., Java 21 or Java 17) unless your course specifies another version.

Install

↓

Next

↓

Next

↓

Finish

---

Check Installation

Open

CMD

Type

`java -version`

Output example

`java version "21"`

Then

`javac -version`

Output

`javac 21`

---

Difference

`java`

↓

Runs Program

`javac`

↓

## Compiles Program

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### First Java Program

```
public class Hello {  
  
    public static void main(String[] args) {  
  
        System.out.println("Hello ");  
  
    }  
  
}
```

Don't worry if it looks scary. We'll break it down.

---

### File Name

Save as

Hello.java

Rule

Class name

=

File name

Hello

↓

Hello.java

If different

Error.

---

## Understanding Every Word

public

Means

Everyone can access.

Think of it like a public park—anyone can enter.

---

class

Class ante

Blueprint.

Real life example:

House build cheyyali ante

First blueprint draw chestaru.

Blueprint

↓

House

Same

Class

↓

Objects

(We'll learn this deeply later.)

---

Hello

Class name.

You can write

Student

Car

Mobile

Calculator

Animal

---

```
public static void main(String[] args)
```

Most beginners fear this line. Let's split it.

```
public
```

Visible to JVM.

```
static
```

JVM object create cheyyakunda direct ga call cheyyagaladu.

```
void
```

This method returns nothing.

```
main
```

Entry point. Program execution starts here.

```
String[] args
```

Stores command-line arguments (we'll learn later).

For now, remember:

**Every standalone Java program starts executing from main().**

---

## Curly Braces {}

Everything inside

```
{
```

```
}
```

belongs together.

Example:

```
{
```

```
    Statement 1
```

Statement 2

}

Think of them as the walls of a room.

---

### **System.out.println()**

This prints output on the screen.

Split it:

System

Java's built-in system class.

out

Standard output (monitor/screen).

println

Prints and then moves the cursor to the next line.

---

Example

```
System.out.println("Hello");
```

```
System.out.println("World");
```

Output

Hello

World

---

### **Difference Between print and println**

```
System.out.print("Hello");
```

```
System.out.print("Java");
```

Output

HelloJava

Because print() does **not** move to the next line.

---

```
System.out.println("Hello");
```



```
System.out.println("Java");
```

Output

Hello

Java

Because println() prints and then goes to the next line.

---

## Compilation Process

You write:

Hello.java

Step 1:

```
javac Hello.java
```

Compiler checks for syntax errors.

If everything is correct:

Hello.class

is created.

.class file contains **bytecode**.

Step 2:

```
java Hello
```

JVM reads Hello.class and executes it.

---

## Complete Flow

You Write Code

↓

Hello.java

↓

javac Hello.java



Hello.class (Bytecode)



JVM



Machine Code



Output

---

### Why Bytecode?

If Java directly created machine code:

- Windows machine code
- Linux machine code
- Mac machine code

would all be different.

Instead, Java creates **bytecode**, and each platform's JVM converts that bytecode into the correct machine code.

That's why Java is **platform independent**.

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### Rules for Java Program

- Only one public class per file.
- File name must match the public class name.
- Java is **case-sensitive**.

Example:

Hello

is **not** the same as

hello

---

## Common Beginner Mistakes

✗ Forgetting a semicolon (;):

```
System.out.println("Hello")
```

Correct:

```
System.out.println("Hello");
```

---

✗ File name mismatch:

```
public class Hello
```

Saved as:

Demo.java

Error!

Must be:

Hello.java

---

✗ Wrong case:

```
system.out.println("Hi");
```

Correct:

```
System.out.println("Hi");
```

---

## Practice Programs

### 1. Print Your Name

```
public class Name {  
    public static void main(String[] args) {  
        System.out.println("Hello");  
    }  
}
```

---

## 2. Print Three Lines

```
public class Intro {  
    public static void main(String[] args) {  
        System.out.println("Name: Hello");  
        System.out.println("College: Aits");  
        System.out.println("Course: EEE");  
    }  
}
```

---

## 3. Difference Between print() and println()

```
public class PrintDemo {  
    public static void main(String[] args) {  
        System.out.print("Java ");  
        System.out.print("Programming");  
  
        System.out.println();  
  
        System.out.println("Hello");  
        System.out.println("World");  
    }  
}
```

Output:

Java Programming

Hello

World

---

## Interview Questions

1. **Who developed Java?**  
James Gosling.
  2. **What is Java?**  
A high-level, object-oriented, platform-independent programming language.
  3. **What is JVM?**  
Java Virtual Machine. It executes Java bytecode.
  4. **What is JRE?**  
Java Runtime Environment. It provides everything needed to run Java programs.
  5. **What is JDK?**  
Java Development Kit. It includes tools to develop and compile Java programs.
  6. **Difference between JDK, JRE, and JVM?**
    - JDK → Develop + Compile + Run
    - JRE → Run programs
    - JVM → Execute bytecode
  7. **Why is Java platform independent?**  
Because Java compiles source code into platform-neutral bytecode, and the JVM on each operating system converts that bytecode into machine code.
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## What You Should Remember Today

By the end of Day 1, you should confidently know:

- ☒ What programming is.
- ☒ Why Java is widely used.
- ☒ Java's history and the "Write Once, Run Anywhere" concept.
- ☒ The difference between **JDK**, **JRE**, and **JVM**.
- ☒ How Java code is compiled and executed.

- ☒ The structure of your first Java program.
- ☒ The purpose of `main()` and `System.out.println()`.
- ☒ The difference between `print()` and `println()`.
- ☒ How to compile (`javac`) and run (`java`) a Java program.
- ☒ The most common beginner mistakes.